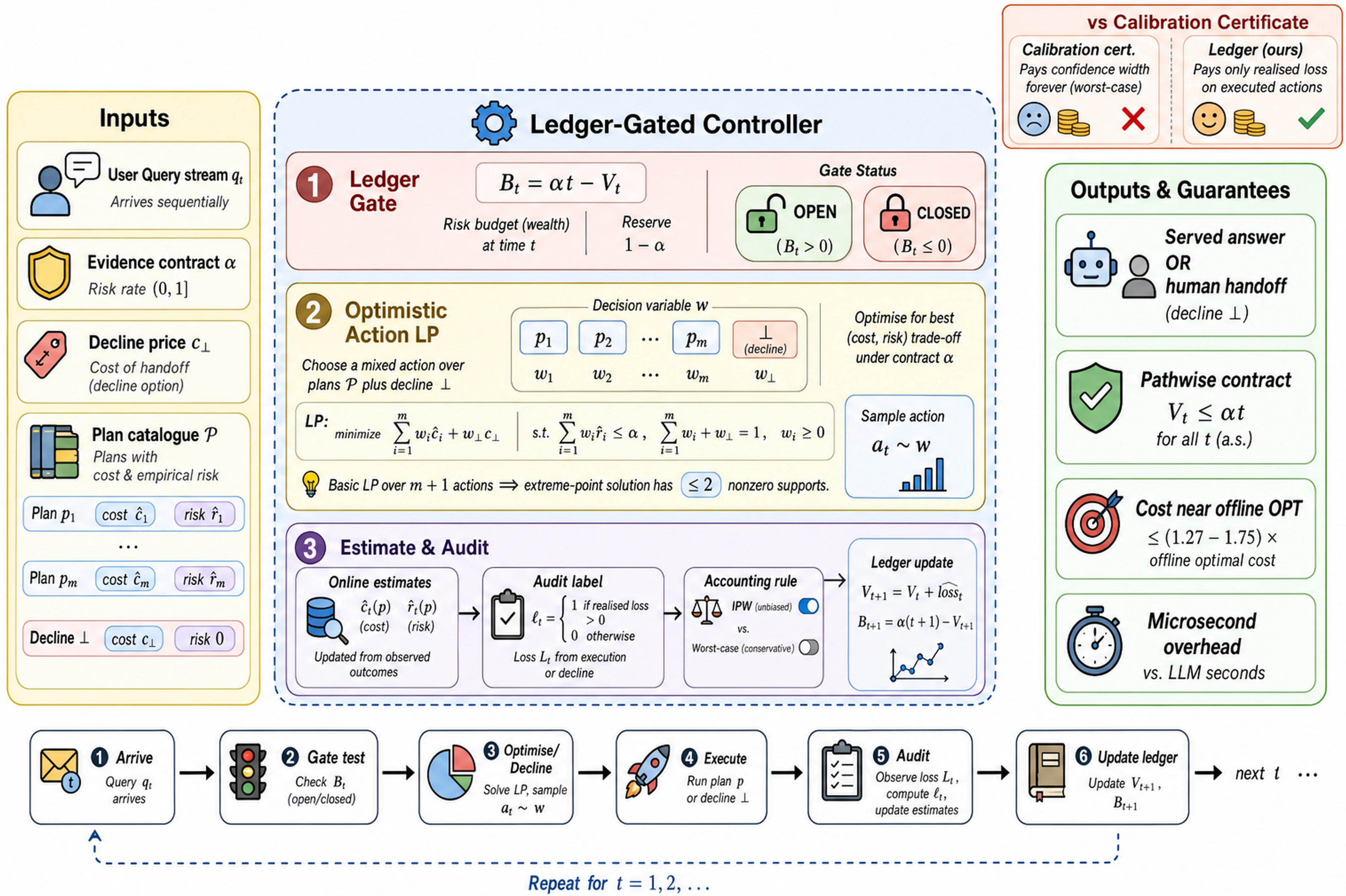

vs Calibration Certificate
Calibration cert.
Pays confidence width forever (worst-case)
Ledger (ours)
Pays only realised loss on executed actions
Inputs
User Query stream q_t
Arrives sequentially
Evidence contract α
Risk rate $(0,1]$
Decline price $c_\perp$
Cost of handoff (decline option)
Plan catalogue $\mathcal{P}$
Plans with cost & empirical risk
Plan p_1 cost $\hat{c}_1$ risk $\hat{r}_1$
...
Plan p_m cost $\hat{c}_m$ risk $\hat{r}_m$
Decline $\perp$ cost $c_\perp$ risk 0
Ledger-Gated Controller
1 Ledger Gate
$B_t = \alpha t - V_t$
Risk budget (wealth) at time t
Reserve $1-\alpha$
Gate Status
OPEN $(B_t > 0)$
CLOSED $(B_t \le 0)$
2 Optimistic Action LP
Choose a mixed action over plans $\mathcal{P}$ plus decline $\perp$
Decision variable w
p_1 p_2 ... p_m $\perp$ (decline)
w_1 w_2 ... w_m $w_\perp$
Optimise for best (cost, risk) trade-off under contract α
LP: minimize $\sum_{i=1}^{m} w_i \hat{c}_i + w_\perp c_\perp$ s.t. $\sum_{i=1}^{m} w_i \hat{r}_i \le \alpha$, $\sum_{i=1}^{m} w_i + w_\perp = 1$, $w_i \ge 0$
Sample action $a_t \sim w$
Basic LP over $m+1$ actions $\Longrightarrow$ extreme-point solution has ≤ 2 nonzero supports.
3 Estimate & Audit
Online estimates
$\hat{c}_t(p)$ (cost) $\hat{r}_t(p)$ (risk)
Updated from observed outcomes
Audit label
$\ell_t = \begin{cases} 1 & \text{if realised loss} > 0 \\ 0 & \text{otherwise} \end{cases}$
Loss L_t from execution or decline
Accounting rule
IPW (unbiased)
vs.
Worst-case (conservative)
Ledger update
$V_{t+1} = V_t + \widehat{loss}_t$
$B_{t+1} = \alpha(t+1) - V_{t+1}$
Outputs & Guarantees
Served answer OR human handoff (decline $\perp$)
Pathwise contract
$V_t \le \alpha t$
for all t (a.s.)
Cost near offline OPT
$\le (1.27 - 1.75) \times$ offline optimal cost
Microsecond overhead
vs. LLM seconds
1 Arrive
Query q_t arrives
2 Gate test
Check B_t (open/closed)
3 Optimise/ Decline
Solve LP, sample $a_t \sim w$
4 Execute
Run plan p or decline $\perp$
5 Audit
Observe loss L_t, compute ℓ_t, update estimates
6 Update ledger
Update V_{t+1}, B_{t+1}
next t ...
Repeat for $t = 1, 2, \ldots$